

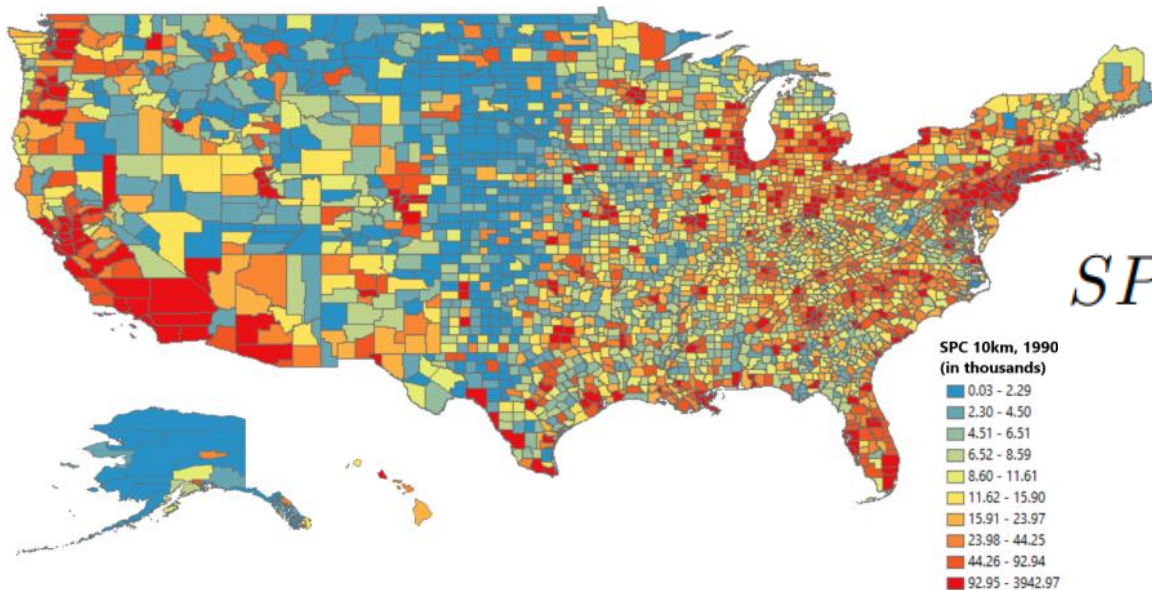
Innovation in social networks

Balázs Lengyel

Economic Complexity, Class 6 – Nov 4 2025

Social Data Science, Corvinus University of Budapest

- ▶ Basic idea: Concentration means less expense on high skilled workers + on product distribution
 - ▶ → Worth to implement new ideas → Population concentration correlates with GDP growth
- ▶ New measurement: Spatial Population Concentration (SPC) → Weighted average population surrounding every person within a geographic area
 - ▶ Better than population density and urbanization
- ▶ USA data from 1975 to 2015. Panel model regression with control variables and fixed effects
- ▶ Increasing the SPC by 1 standard deviation (in 25km radius), ceteris paribus GDP growth will be greater by 7.63%



$$SPC_d = \frac{\sum_{i=1}^N pop_i * n_{di}}{\sum_{i=1}^N pop_i} - 1$$

Zoltán Cséfalvay. (2024): Europe's Scaleup Geography and the Role of Access to Talent

How does access to locally available talent shape geography of European scaleups?

13,851 European scaleups mapped to 166 EU-OECD FUAs

K-means clustering

Talent proxies

Simple linear regression

Top hubs capture a **disproportionate share of outcomes** ($\approx 61\%$ of all scaleups, 71% of funding, 68% of €200m+ firms concentrated in ~ 15 cities).

Clear **West-East** and **North-South** gradient, so national capitals in Western/Northern Europe far outpace peers. cities plugged into strong cross-border networks benefit most; peripherals **depend on proximity and links** to nearby hubs..

“Upstream” talent and “downstream” venture outcomes results in cities that **attract, train, and retain students** generate larger founder pools and more scaleups.

Goal: Strengthening of university-startup bridges

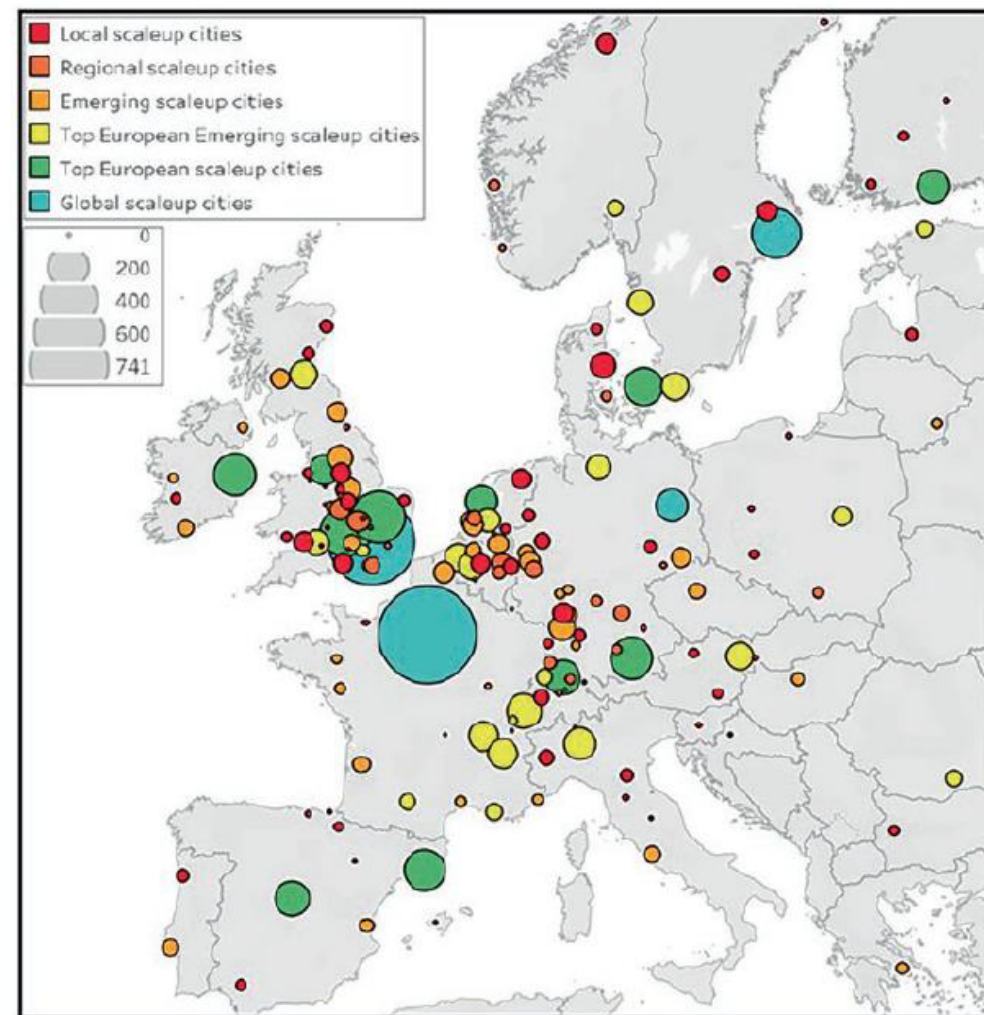
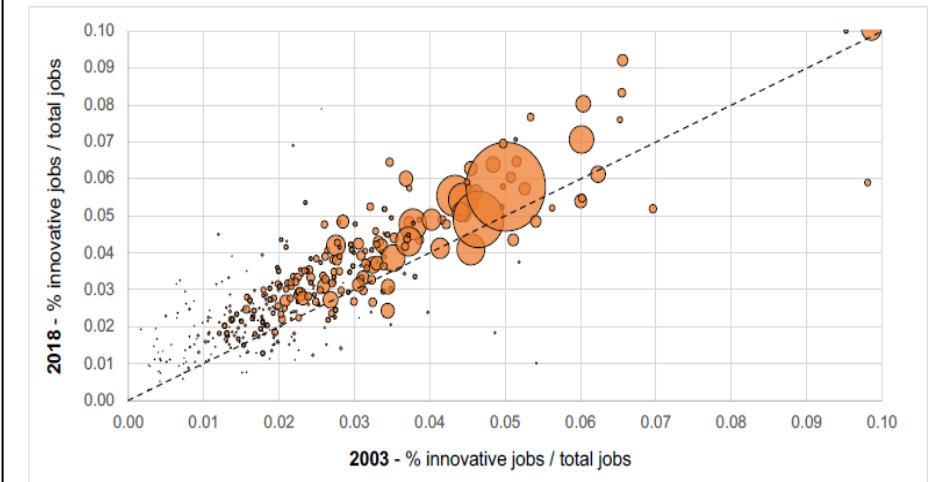


Fig. 6.9 Number of founders attended a university of the scaleup city who raised more than €10 million in funding. Europe, 2021. Source: Author's own calculation based on Dealroom data retrieved June 2, 2021 from <https://dealroom.co> and EU-OECD FUA classification. Design by author

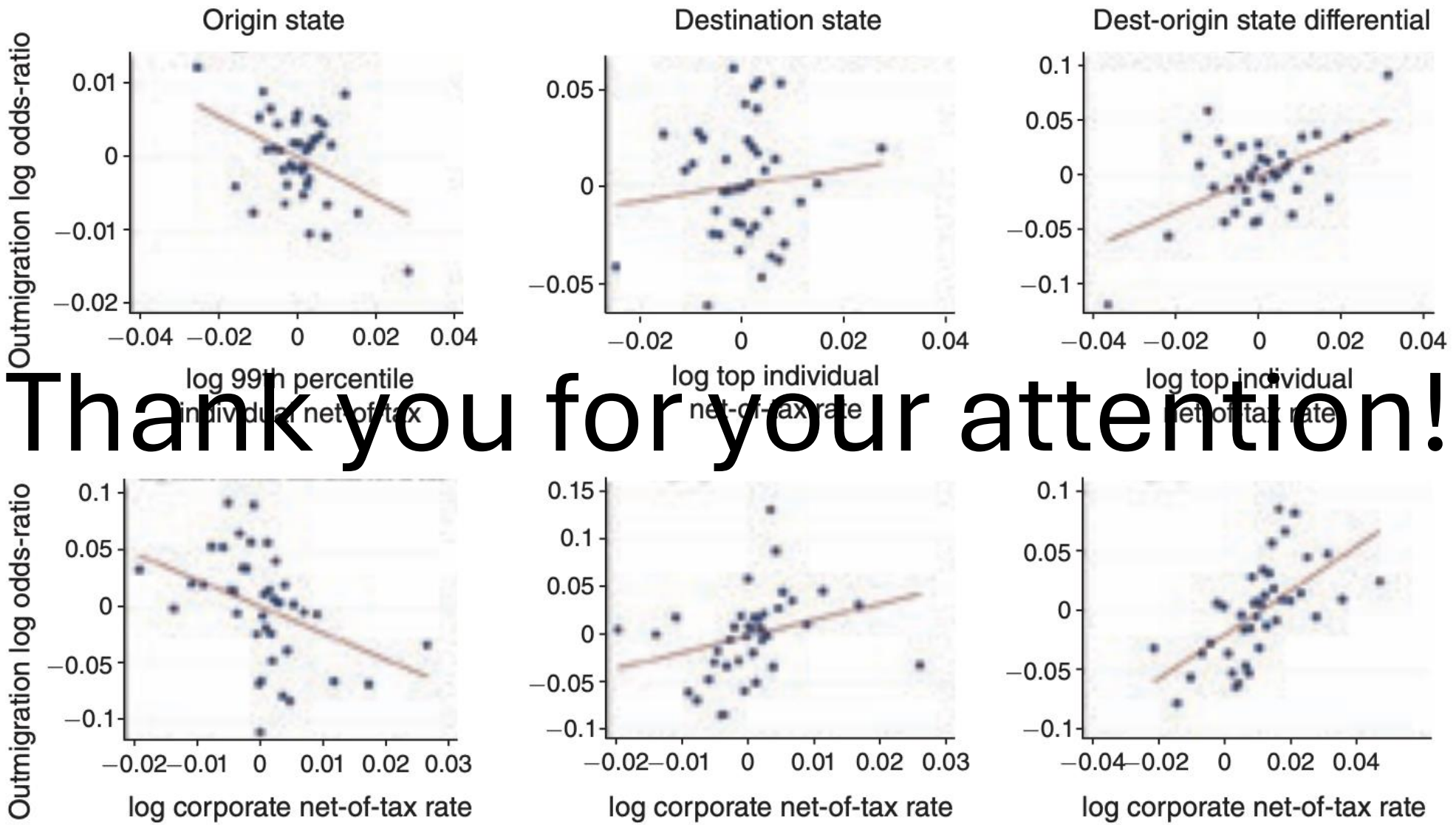
Milene Simone Tessarin & Carlos Roberto Azzoni (2023), Innovation and Competitiveness: The Regional Dimension

- Research question and focus
 - How does labour pools and geographic concentration influence regional innovation and the pace of technological change
 - Developing economies → human capital and workers skills → innovative potential outside R&D
- Methodology & Data
 - Unit of analysis: 374 labour market areas (LMA) across Brasil
 - Data: from RAIS → from 2003 to 2018 50 million formal workers
 - Measuring innovative potential by occupational skills → Six innovative skills via PCA: creativity, active learning, design, engineering/technology, technology design, updating knowledge → Innovation potential index = weighted sum of innovative occupations
 - Empirical model panel regression 2003-2018 4 non-overlapping 4 year periods → to test for convergence or divergence across regions
- Results
 - High concentration: 44 LMAs (12%) hold more then 80% of innovative jobs → close to 20-80 rule
 - Lagging regions grew faster → slow convergence (evidence from the regression as well)
 - Richeher regions are more innovative, industrial structure drives innovation, large firms are more likely to innovate → Schumpeter scale effect
- Conclusion
 - Regional path dependence matters, BUT redistributive policies helped the peripheries to catch up, still a big difference
 - Modest deconcentration of innovative potential unlike Europe
 - Still remains geographically concentrated

Figure 5 – Change in the share of innovative jobs by regions – Selected group (2003-2018)

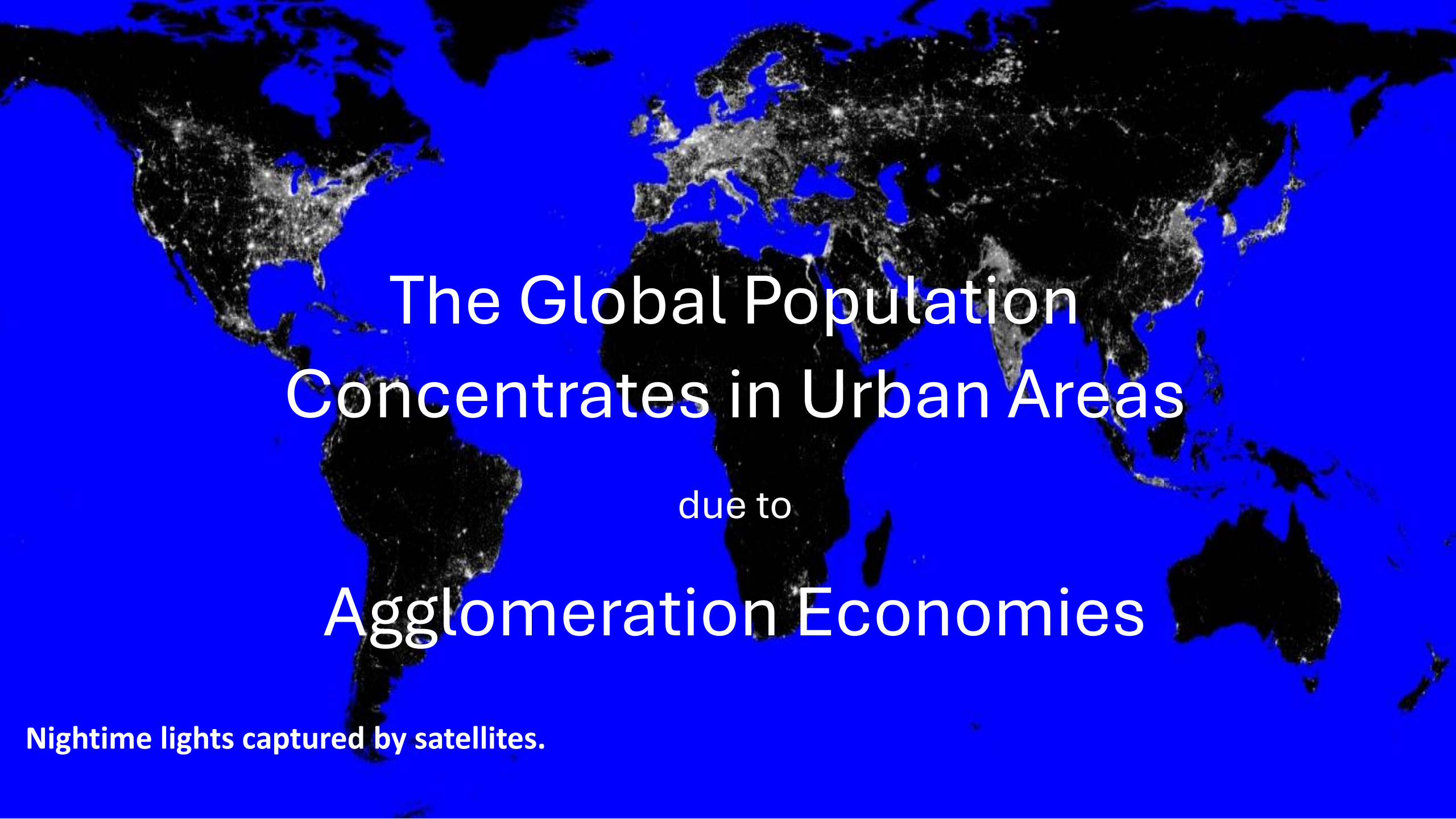


Source: Author's elaboration. Note: The bubble size represents the number of innovative jobs by region in 2003. Includes only regions with up to 10% of innovative jobs.



Thank you for your attention!

FIGURE 4. OUTMIGRATION VERSUS NET-OF-TAX RATES

A world map with a black background, where landmasses are outlined in white. The map is filled with numerous small, bright white dots and clusters of dots, representing nighttime lights captured by satellites. These lights are most concentrated in North America, Europe, and East Asia, with smaller clusters in South America, Africa, and Australia. The text is overlaid on the map.

The Global Population Concentrates in Urban Areas

due to

Agglomeration Economies

Nighttime lights captured by satellites.

We live in a small world



facebook

Diffusion of viruses and information is fast





The global society is segregated

Why do we observe growing inequalities if diffusion of ideas and knowledge in social networks is very quick?

Prosperous areas next to a slum in Mumbai.

Network fragmentation and inequalities

Small-world social networks are fragmented by groups.

One of the strongest forces of group formation is prosperity.

Diffusion of innovation is unequal across network fragments.



The second block

3	Oct 7	Balázs Lengyel Knowledge, innovation, and technological development	II. Innovation and Economic Geography	[4]
4	Oct 14	Balázs Lengyel The development of cities and regions: agglomeration economies, the specialization-diversity debate		[5,6]
5	Oct 21	Zoltán Elekes Labor markets and skills: labor mobility, skill-relatedness		[7,8]
6	Nov 4	Balázs Lengyel Learning and innovation in networks: knowledge sourcing and dynamics of networks		[9,10]
7	Nov 11	Balázs Lengyel and Sándor Juhász Spatial and network inequalities: social capital, segregation and mixing		[11,12]
8	Nov 18	Presentation of students' research proposal		

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[7] Neffke, F., Henning, M., & Boschma, R. (2011). How do regions diversify over time? Industry relatedness and the development of new growth paths in regions. *Economic Geography*, 87(3), 237-265.

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[10] Abbasiharofteh, M., Kogler, D. F., & Lengyel, B. (2023). Atypical combinations of technologies in regional co-inventor networks. *Research Policy*, 52(10), 104886.

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[12] Aiello, L. M., Vybornova, A., Juhász, S., Szell, M., & Bokányi, E. (2025). Urban highways are barriers to social ties. *PNAS*, 122(10), e2408937122.

Today

1. Small-world networks
2. Benefits in small-world networks
3. Geography of social and collaboration networks
4. Innovation in small-world networks

1. Small-world networks



Frigyes Karinthy

Chains

SZÉPIRODALMI  KÖNYVKIADÓ

1929

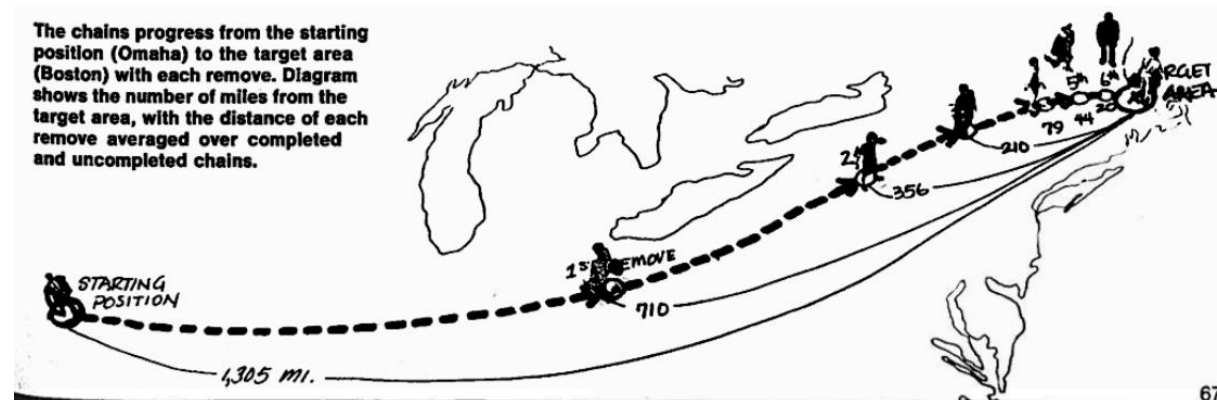
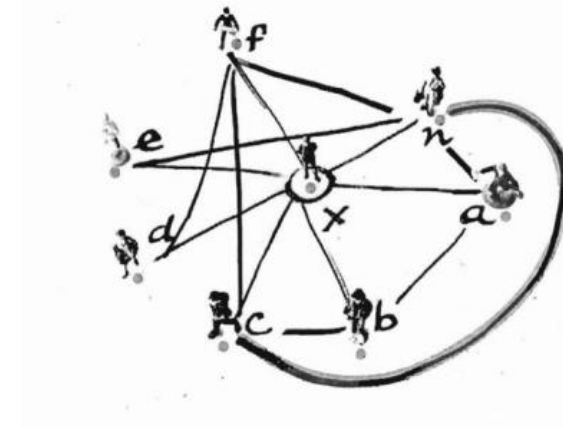
Small worlds

A fascinating game grew out of this discussion. One of us suggested performing the following experiment to prove that the population of the Earth is closer together now than they have ever been before. We should select any person from the 1.5 billion inhabitants of the Earth—anyone, anywhere at all. He bet us that, using no more than five individuals, one of whom is a personal acquaintance, he could contact the selected individual using nothing except the network of personal acquaintances. For example, “Look, you know Mr. X.Y., please ask him to contact his friend Mr. Q.Z., whom he knows, and so forth.”

“An interesting idea!” someone said “Let’s give it a try. How would you contact Selma Lagerlöf?”



Small-world networks

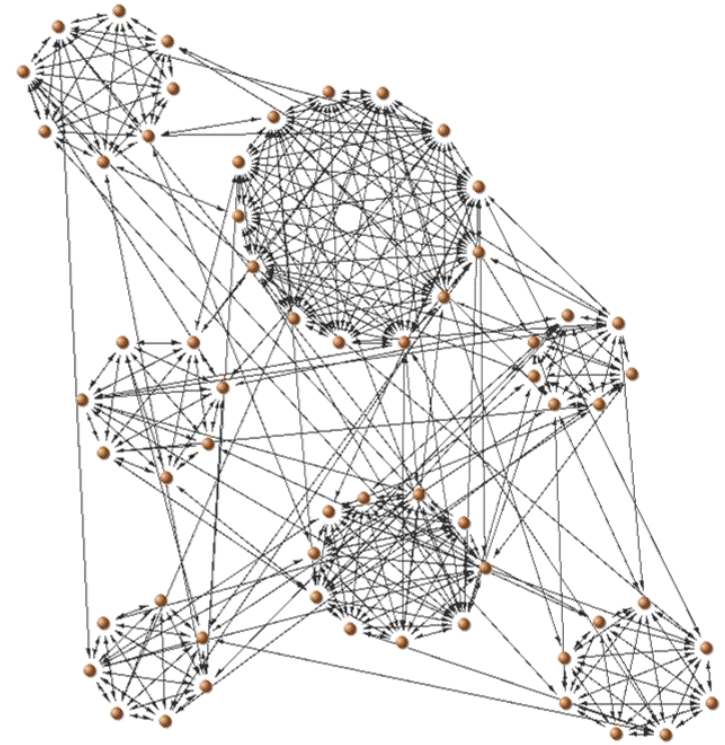


Milgram, S. (1967) The Small-world Problem. *Psychology Today*, 1, 1, 61-67.

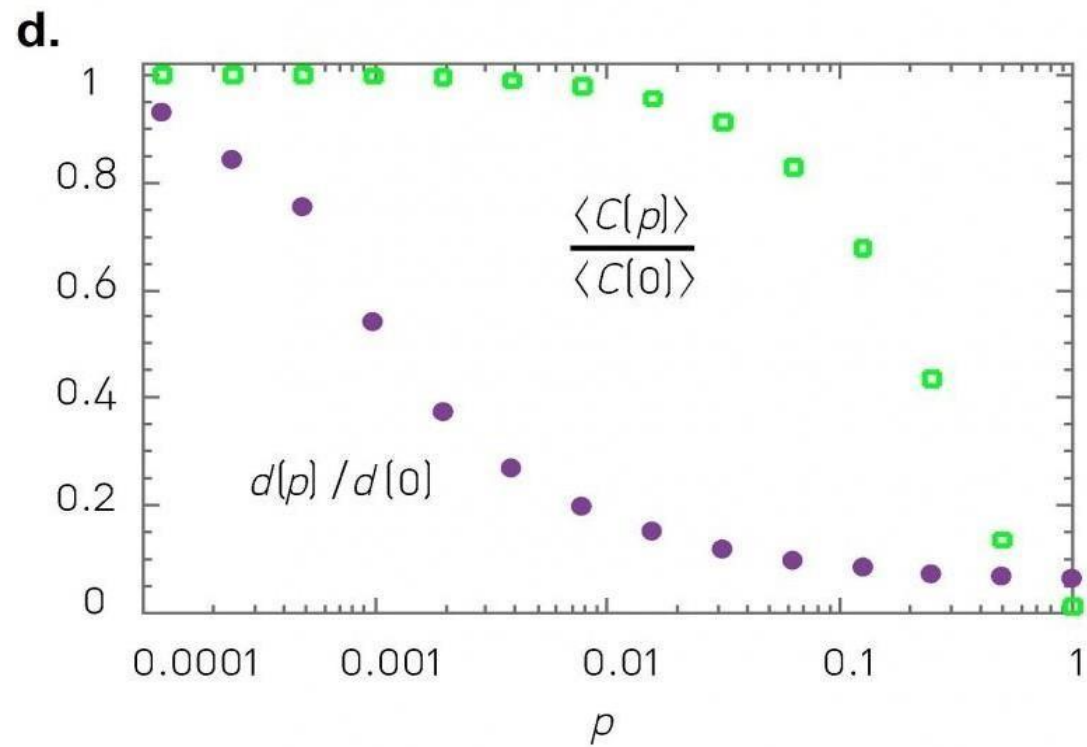
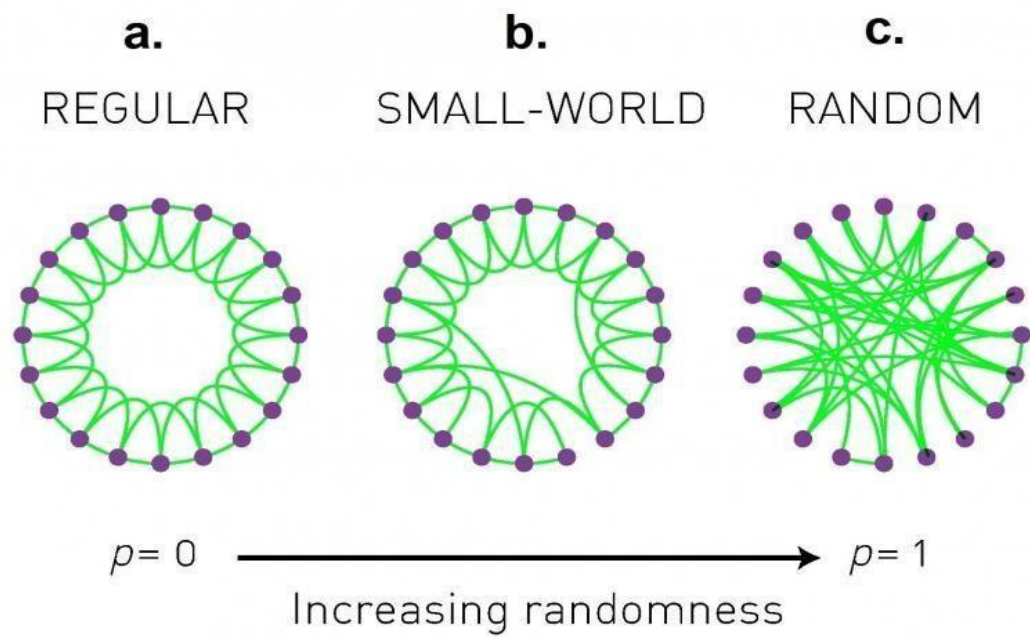


Small-world networks

1. Dense local relations / people share many friends
2. Few bridges make paths short

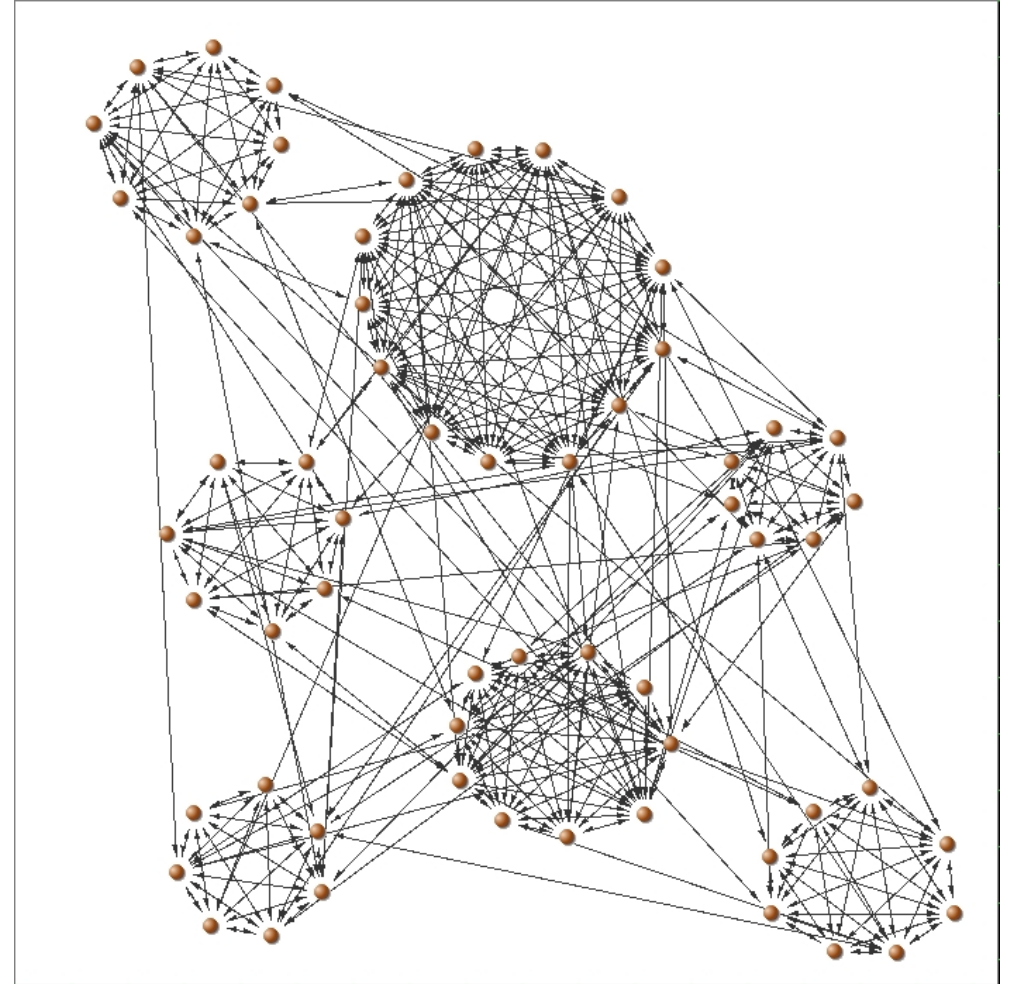


Watts, D.J., Strogatz, S.H. (1998) Collective Dynamics of Small-world Networks. *Nature*, 393, 440-442.



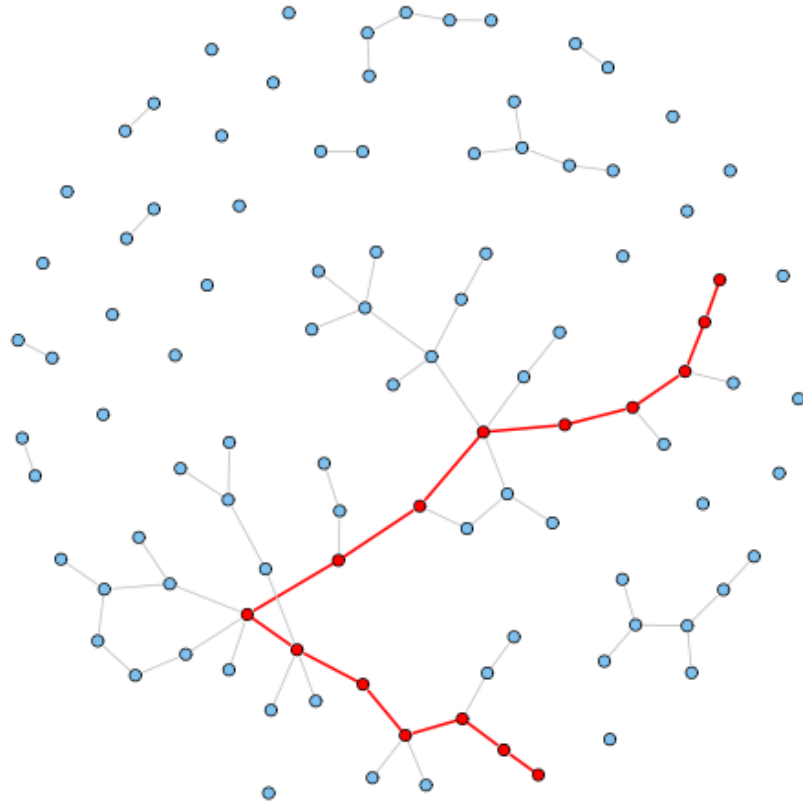
Small worlds

- Social networks cannot be described by random ties:
Dense local relations and
few bridges between isolated groups
- Small worlds
 - High clustering – due to dense local networks
 - Short paths – due to bridges



2. Benefit in small-world networks

Path



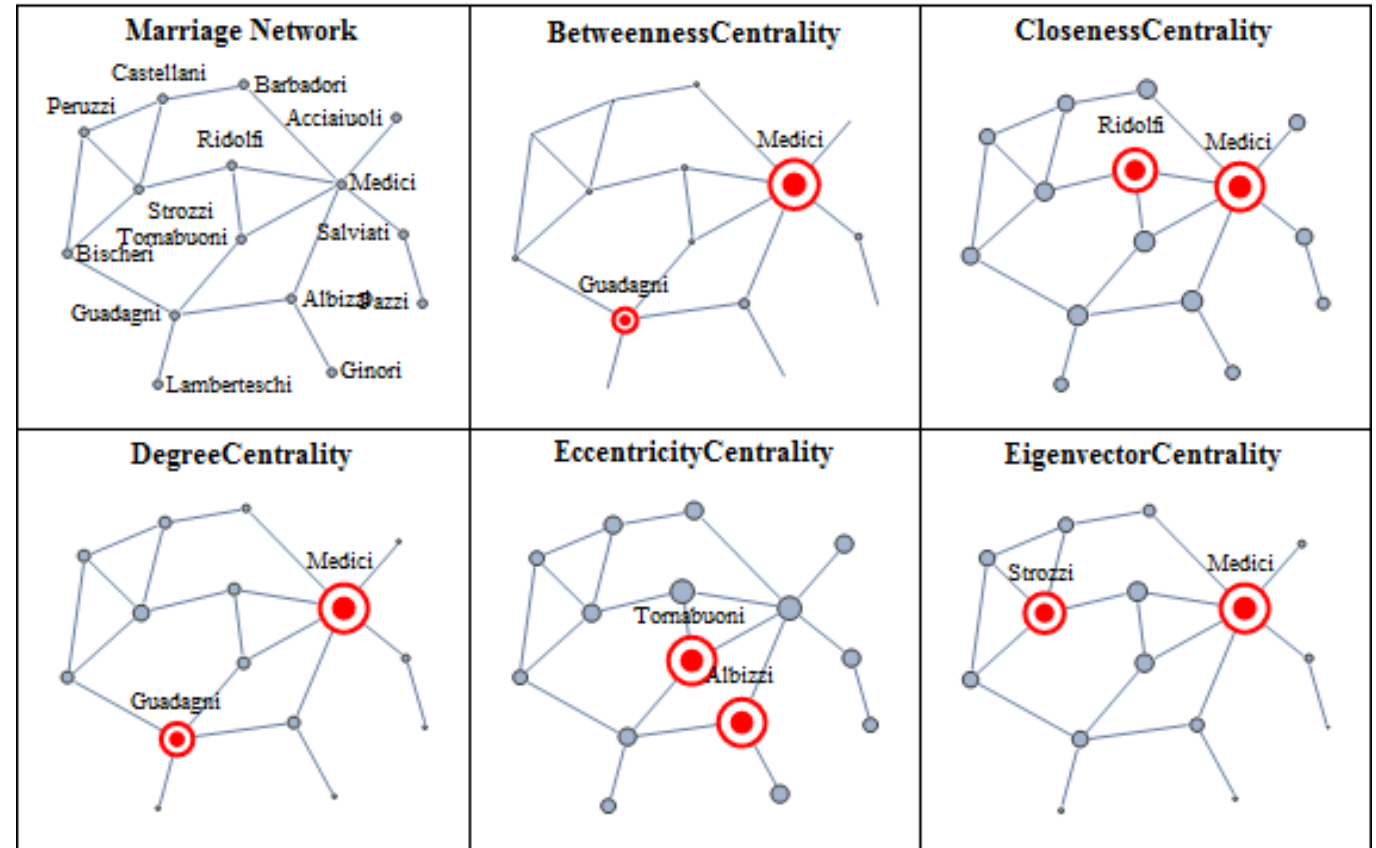
- Path: The set of edges that we can get from one node to any other node
- Distance: the number of steps needed to get from one node to another in the network
- Shortest path: the path between two nodes that includes the fewest steps

Network features: centrality of nodes

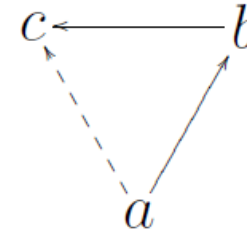
Degree centrality: The number of connections of nodes. For directed networks, we distinguish indegree and outdegree.

Betweenness centrality: the number of shortest paths that pass through the vertex.

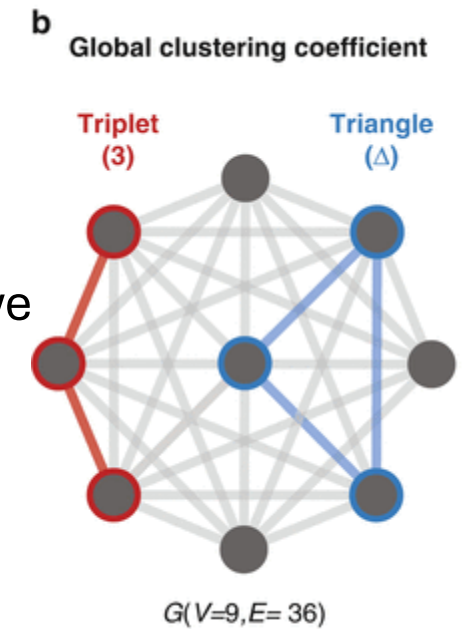
Closeness centrality: the inverse of the sum of distances from the node to all other nodes.



Clustering

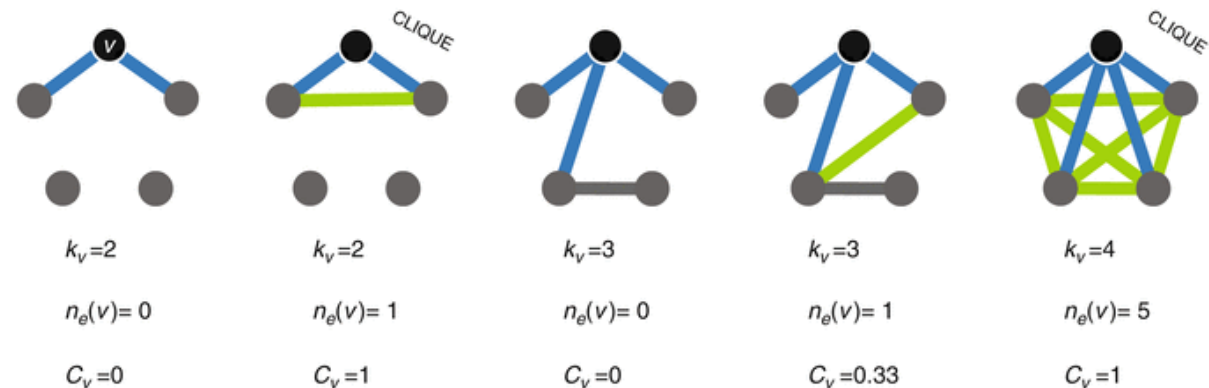


- Transitivity: if 'a' knows 'b' and 'b' knows 'c', then 'a' is likely to know 'c' as we
 - Note that we concentrate on triadic relations in this measure!
- Global clustering: how many triads are closed out of all possible triads?
- Local clustering: for all j, k neighbor-pairs of l , *how many times does the $j-k$ edge materialize?*



(one can sum this up
for the full network)

a Local clustering coefficient



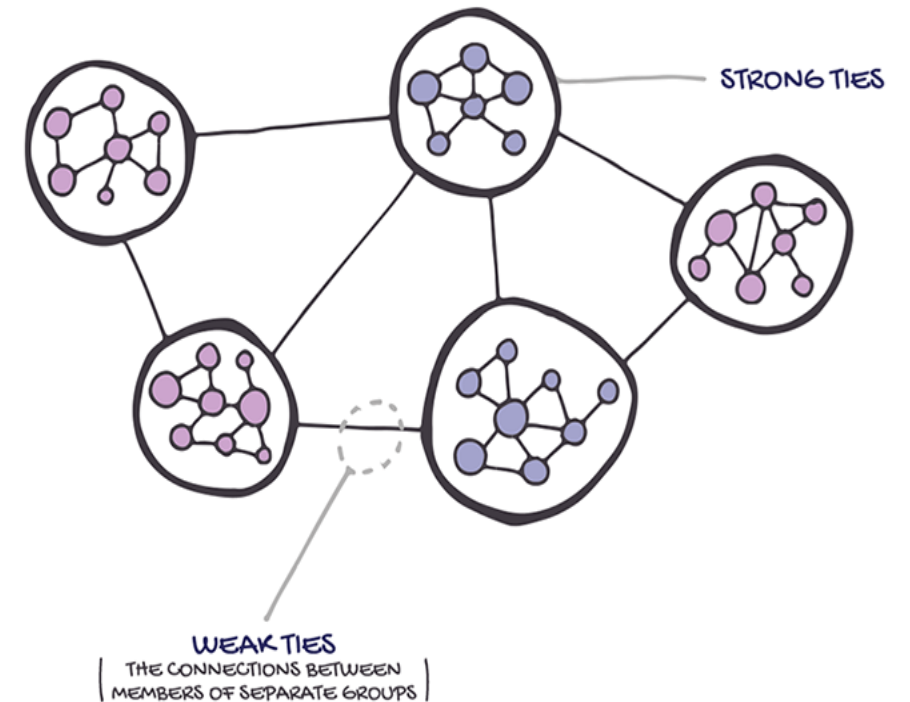
Weak ties and Brokers

Weak ties (Granovetter, 1973):

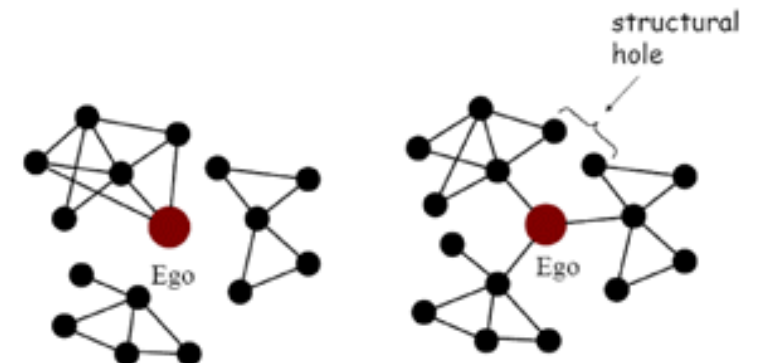
- Valuable information comes from those contacts who are in occasional/not frequent relation with us.

Advantage of Brokers (Burt, 1992)

- Information is homogenous and redundant in dense clusters.
- In case information is not overlapping between two loosely knit clusters, there is a structural hole between these clusters.
- Brokers:
 - tertius gaudens -> control of flows
 - tertius iungens -> establish links



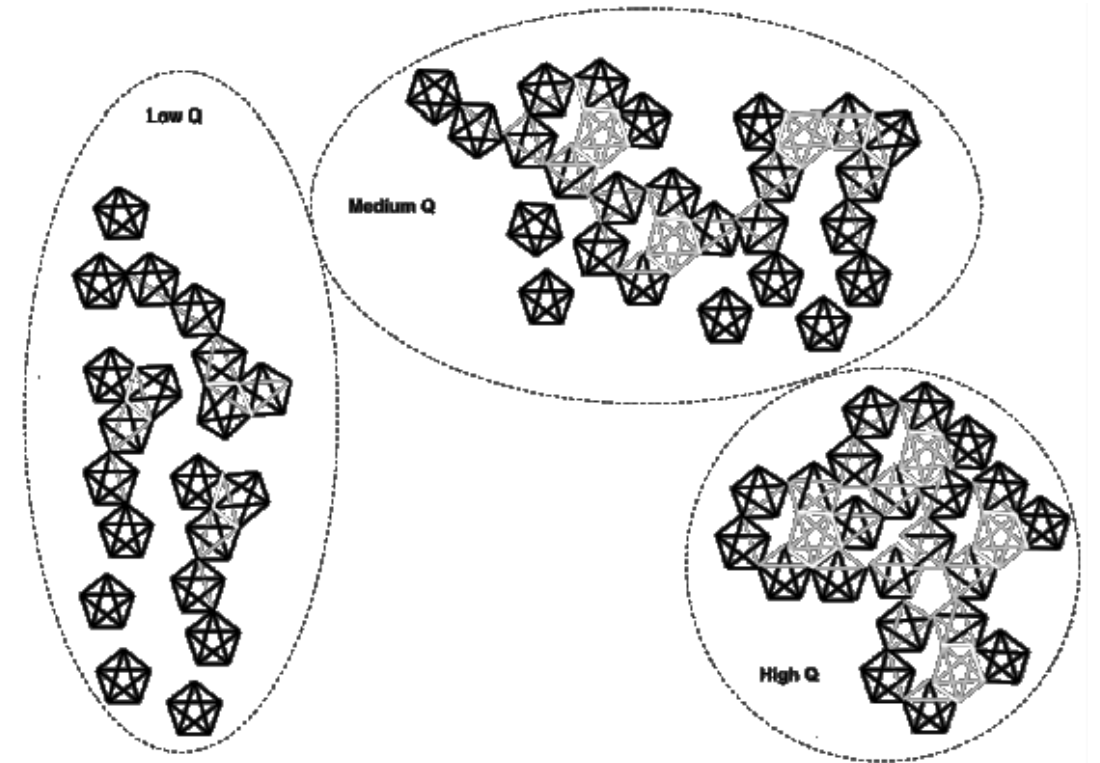
Structural Holes



Optimal small worldliness: combination of diverse and cohesive networks

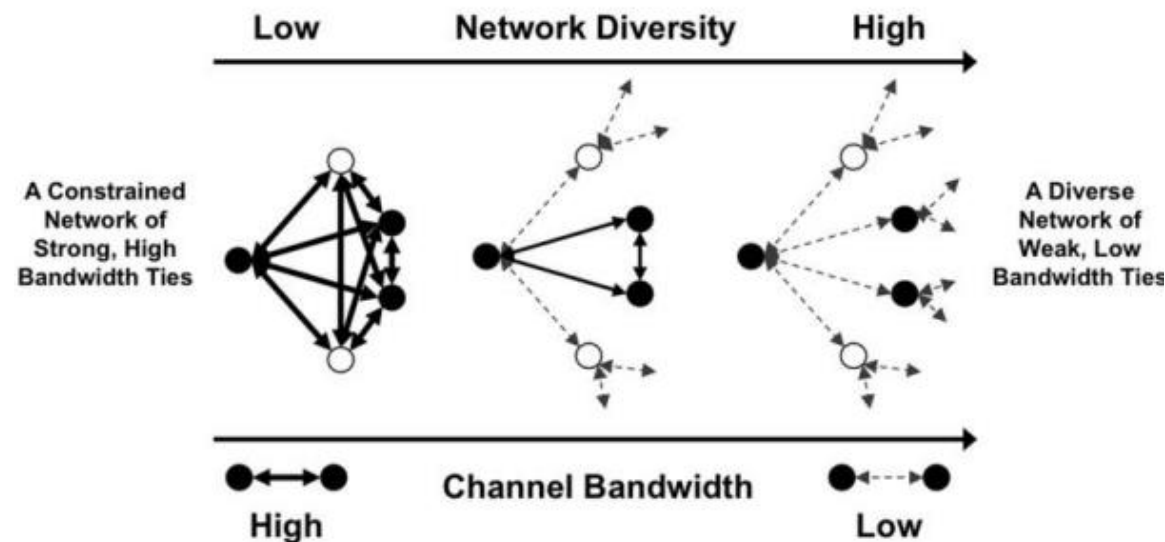
Uzzi and Spiro (2005) American Journal of Sociology

- Small-world Q : high transitivity and short paths
- Broadway musical example: high quality performance needs medium small-worldliness



Benefits in small-world networks

- Cliques / strong ties help us to develop trust, and process information (Coleman, 1988)
- Bridges / weak ties provide us with diverse information (Granovetter, 1973; Burt, 1992)
- A combination of them are needed for success (Uzzi and Spiro, 2005)



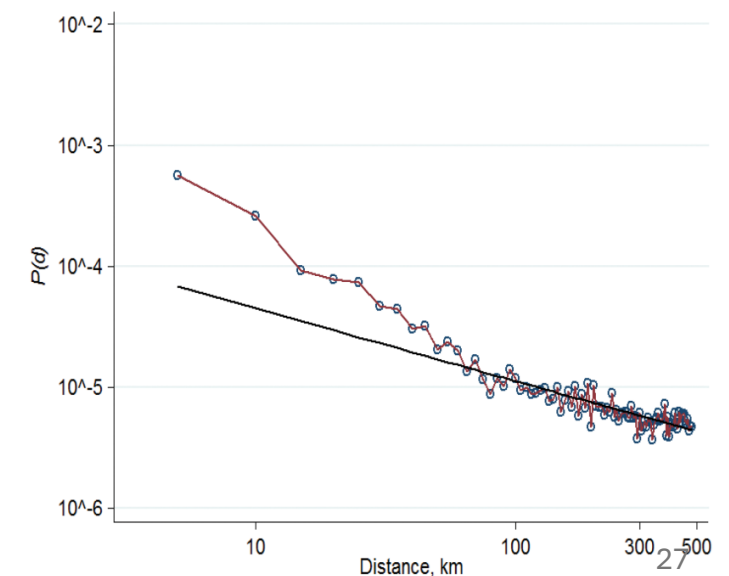
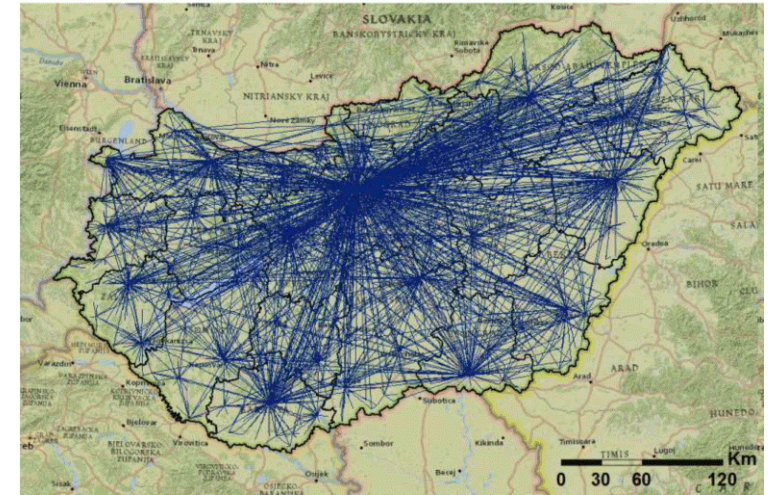
3. Geography of social and collaboration networks

Distance effect

- Creation and maintenance come with costs
- Costs increase with distance
- Gravity models help us analyze the distance-cost relation

$$P = \sum_d L_{ij} / \sum_d N_i \times N_j$$

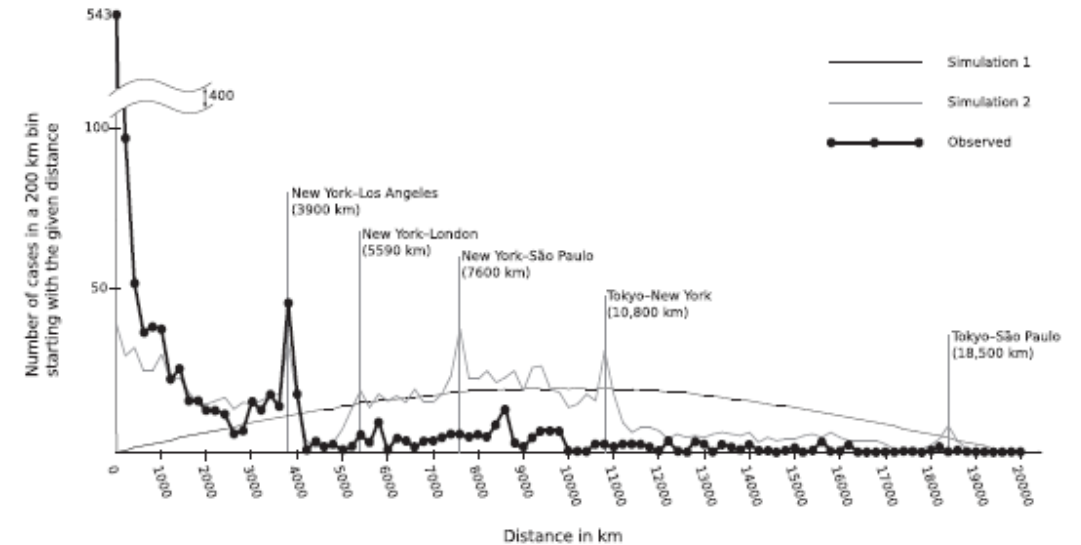
- The probability of triadic closure decreases with distance



Liben-Nowell et al. (2005) PNAS; Lambiotte et al. (2008) Physica A; Lengyel et al (2015) PLOS ONE

Spatial barriers

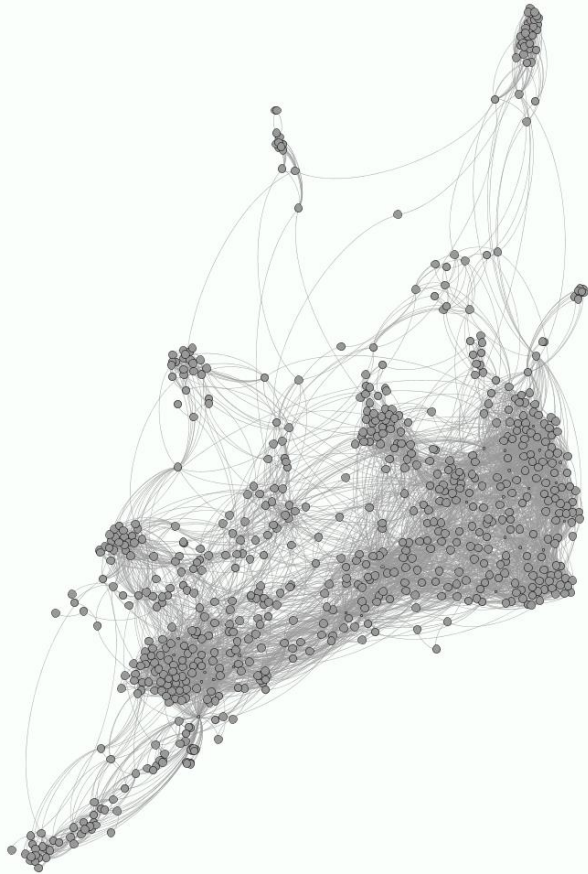
- Spatial mobility determines the geography of social networks
- Spatial barriers hinder mobility and influence network structure



Modularity

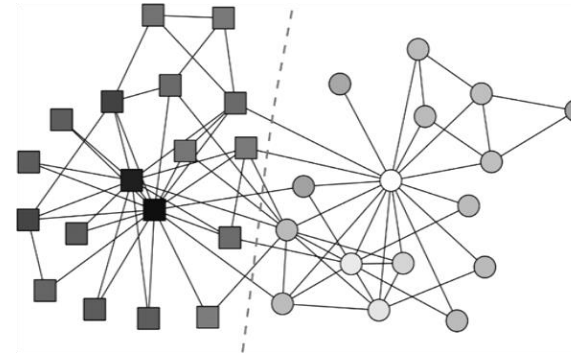
Moduls

Groups/subnetworks/moduls/communities emerge due to transitivity



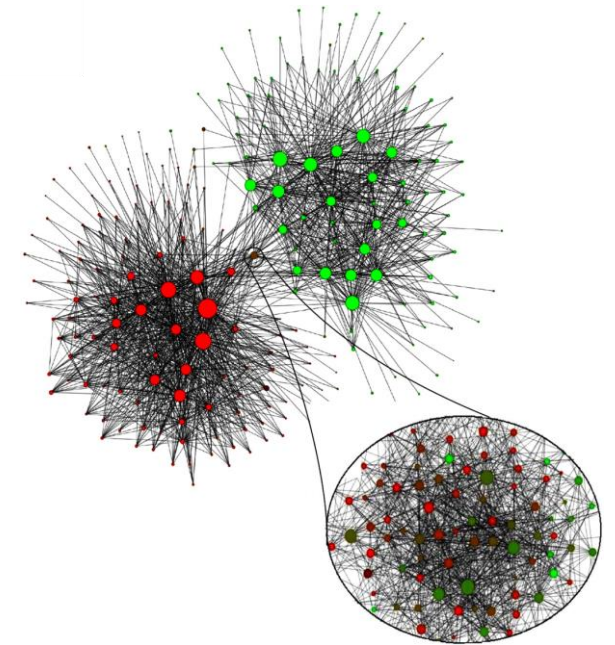
Density Hypothesis of Community detection:

Communities correspond to locally dense neighborhoods of a network



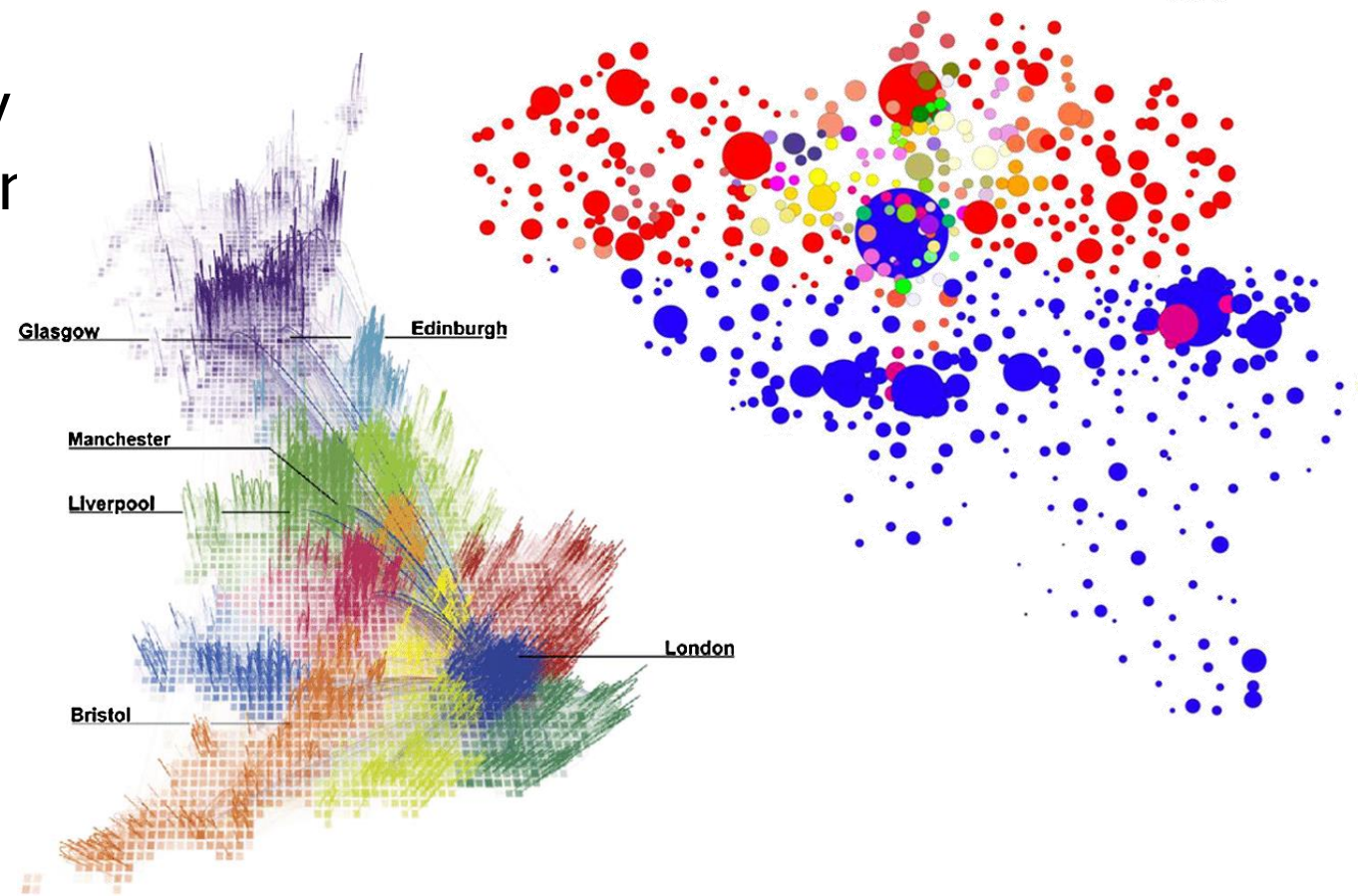
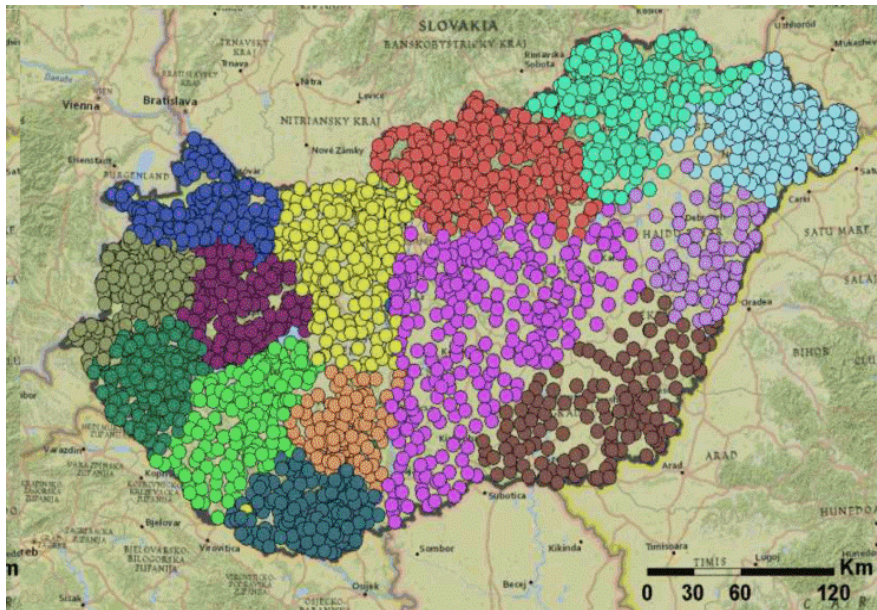
→ Karate Club:
Breakup of the club

→ Belgian Phone Data:
Language spoken

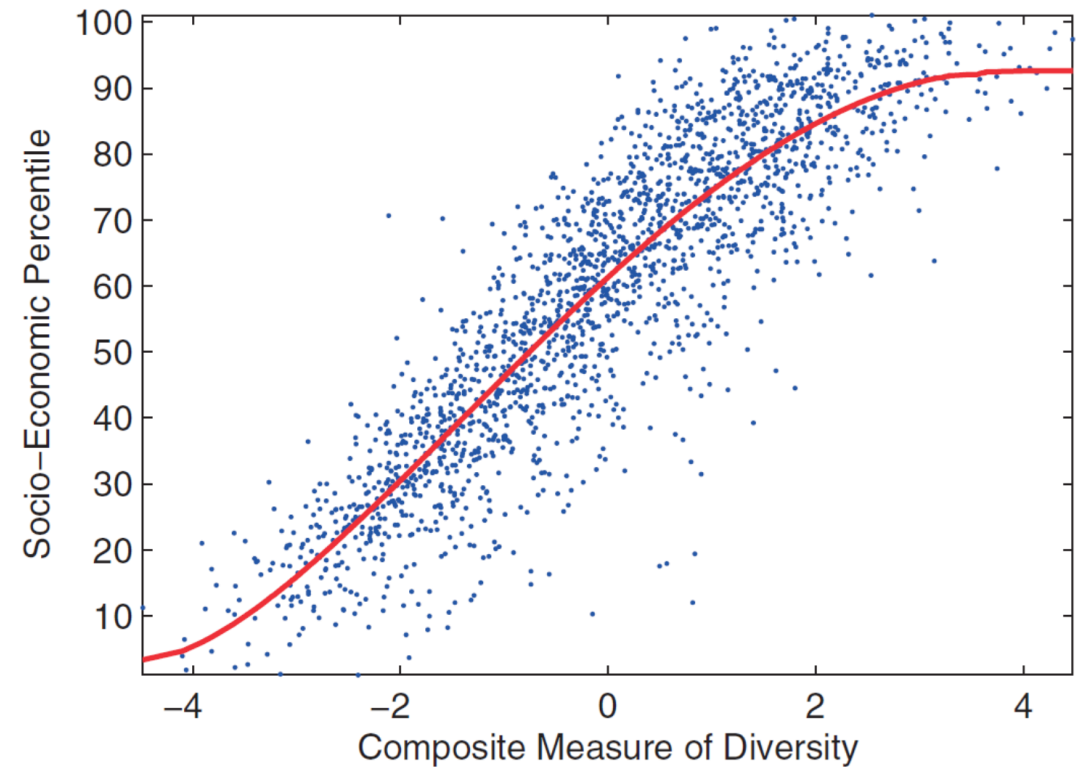
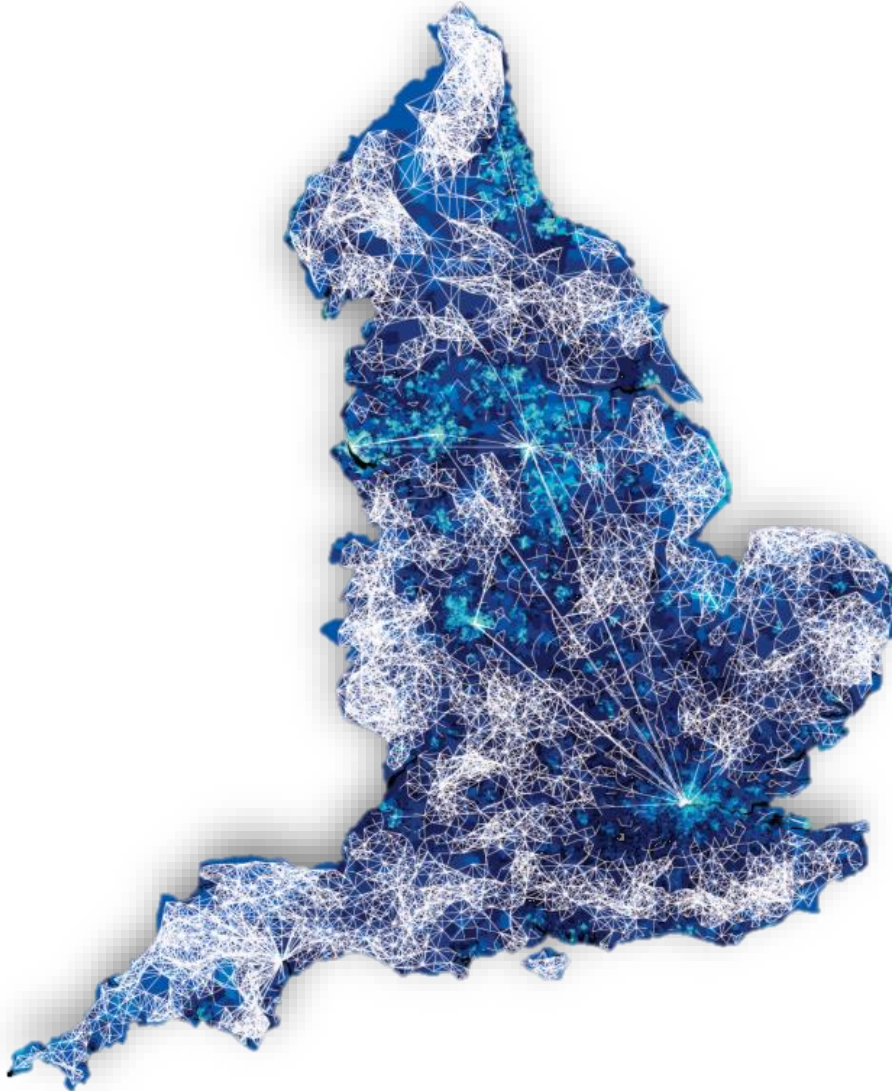


Spatial modules

- Social networks break down by administrative or cultural region



Social networks, geography and wealth

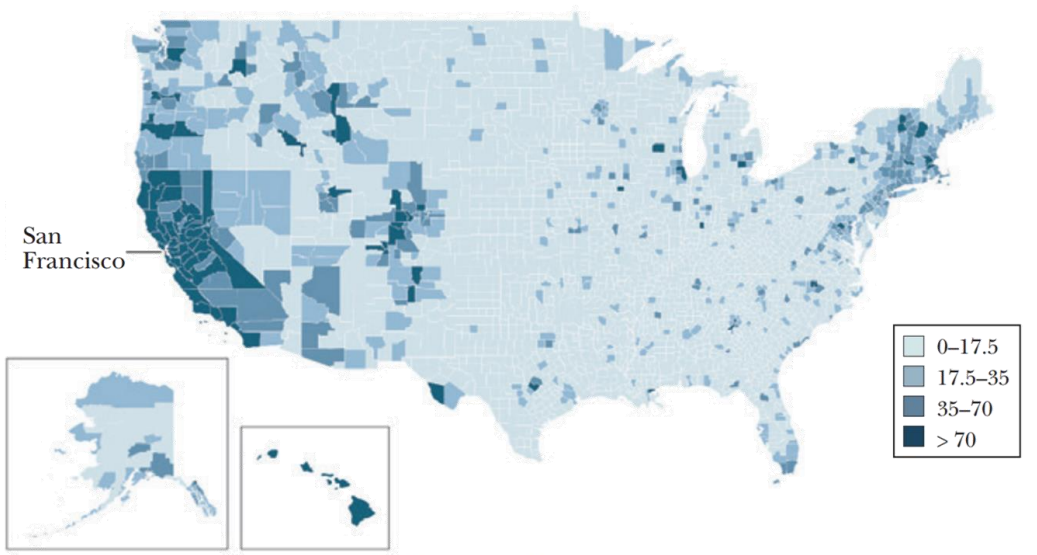


Eagle-Macy-Claxton (2010) Science

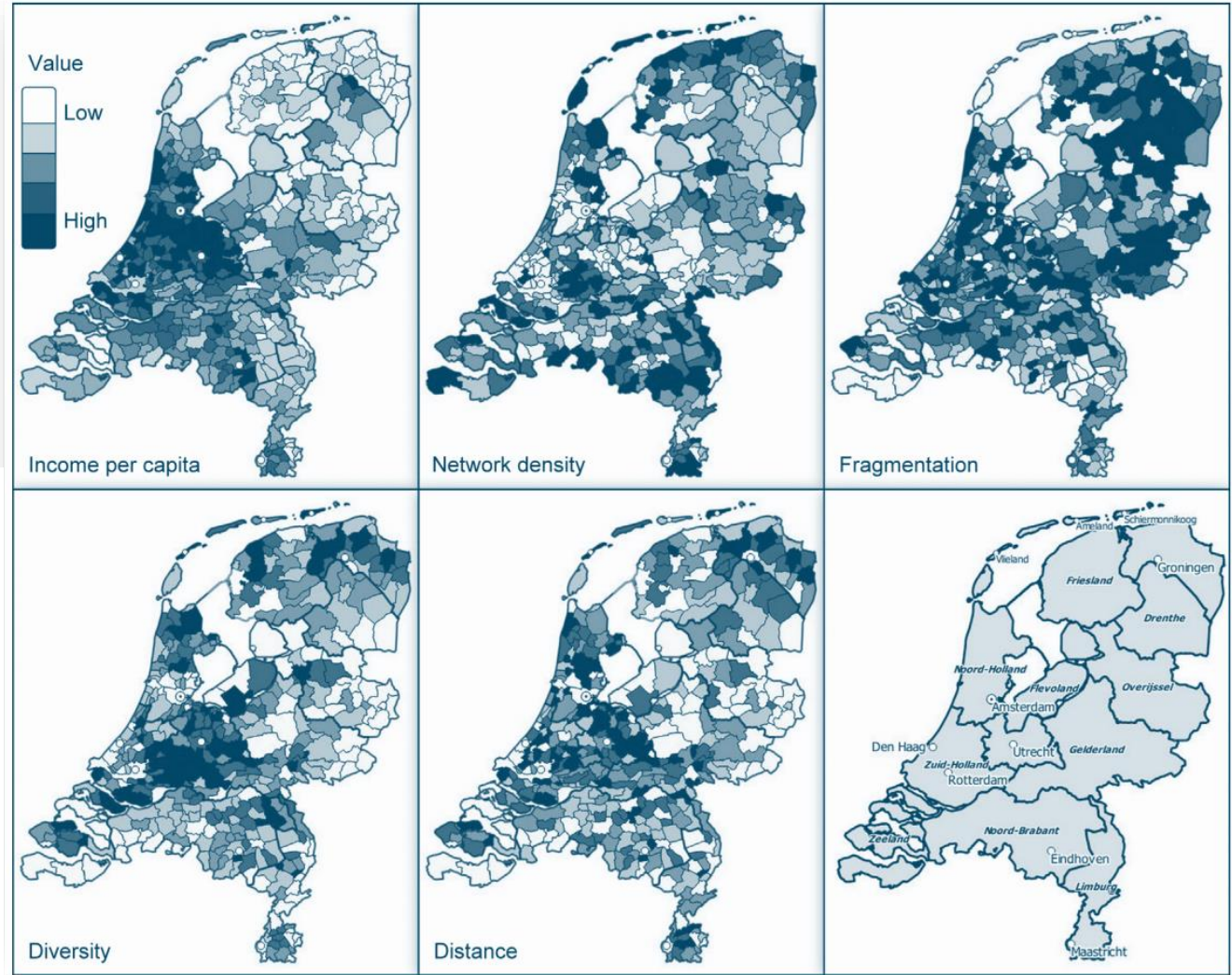
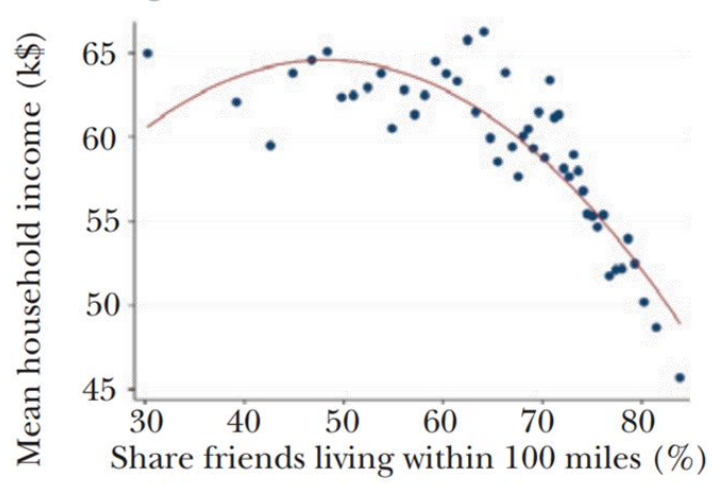


Aggregate social network structures correlate with regional average of individual income

A: Relative Probability of Friendship Link to San Francisco County, CA



A: Average Income



4. Innovation in small-world networks



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Atypical combinations of technologies in regional co-inventor networks

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^c Department of Structural Change, Leibniz Institute of Agricultural Development in Transition Economies, 06120 Halle, Germany

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Small Worlds and Regional Innovation

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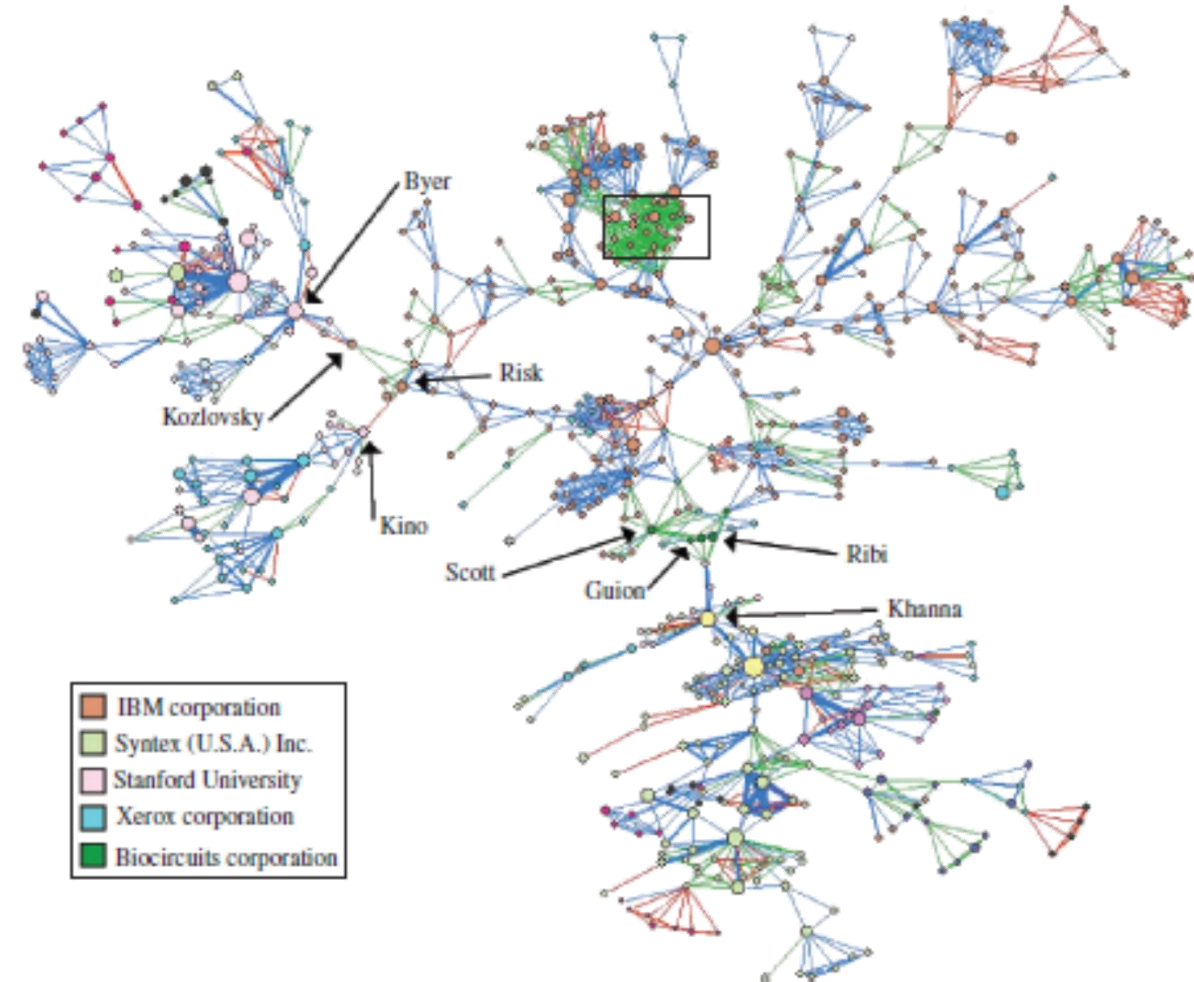
Charles King III

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Short access in the co-inventor network speeds up information flow and induces the combination of knowledge.



Homophily / assortativity

complete heterophily

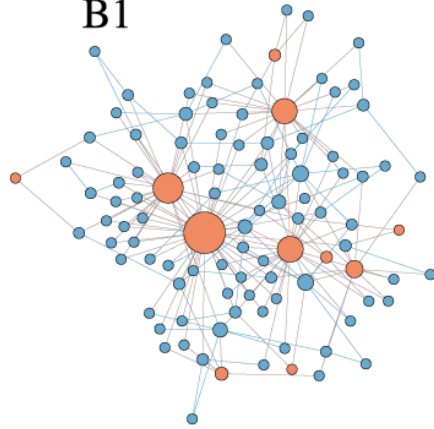
$h = 0$

A1



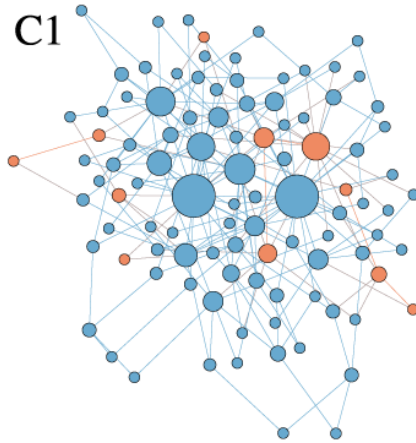
$h = 0.2$

B1



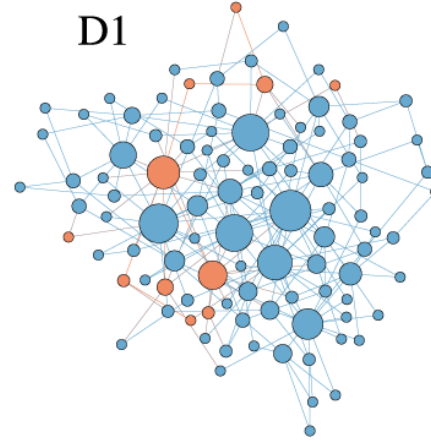
$h = 0.5$

C1



$h = 0.8$

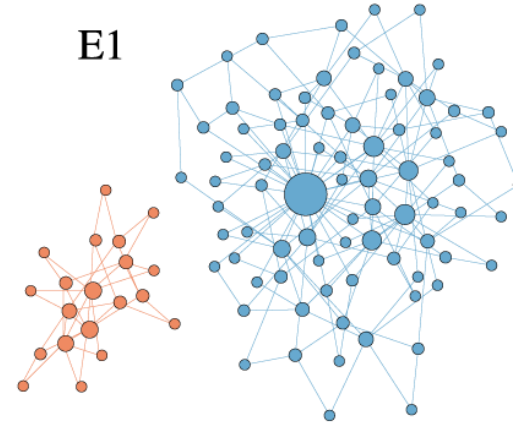
D1



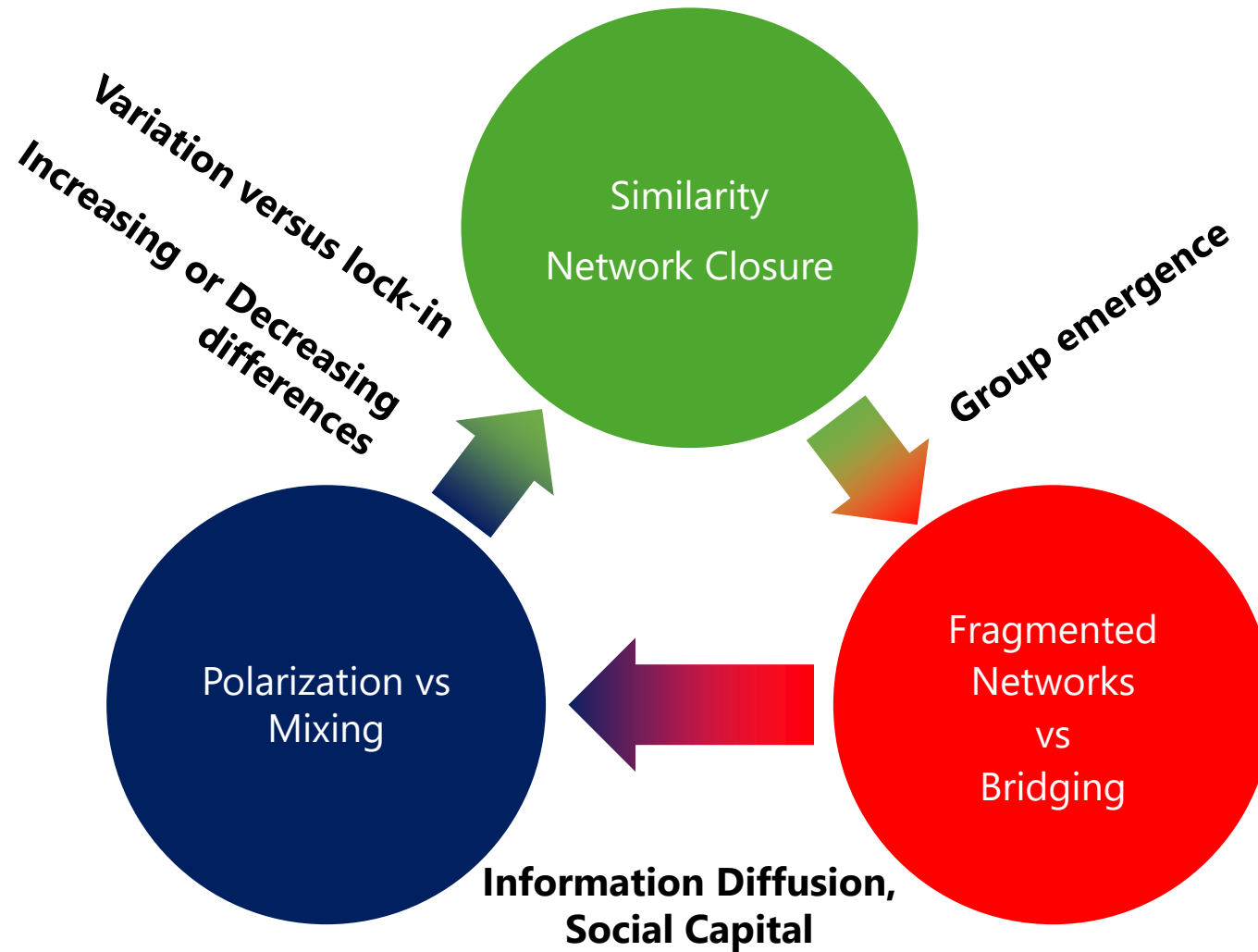
complete homophily

$h = 1$

E1



A Network Dynamics Framework



The role of knowledge: diversity vs specialization

Radical novelty needs diverse knowledge (Florida et al. 2017, Glaeser et al. 1992).

Specialization can also facilitate innovation (Beaudry and Schiffauerova, 2009; Lobo and Strumsky, 2008) when critical masses of experts specialized in distinct pieces of knowledge (Castaldi et al., 2015) are connected through knowledge transfer mechanisms (Berkes and Gaetani, 2020).

Boston biotechnology example (Cooke 2002, Powell et al. 1996): distinct local critical masses in engineering and biology connected by social ties.

The role of knowledge: network dynamics

Similar specialization in knowledge increases link probability.

- Local collaboration networks can become too cohesive and locked-in into technologies – Grabher (1993), Boschma & Frenken (2010), Giuliani (2013), Balland et al. (2016)

Inter-regional links can introduce novelty

- Inter-regional links combined with dense local networks
 - theory: Bathelt et al. (2004), Glückler (2007);
 - empirical evidence: Breschi and Lenzi (2016), Eriksson and Lengyel (2019); Kogler et al. (2023)
- Local variation – Juhász and Lengyel (2018)

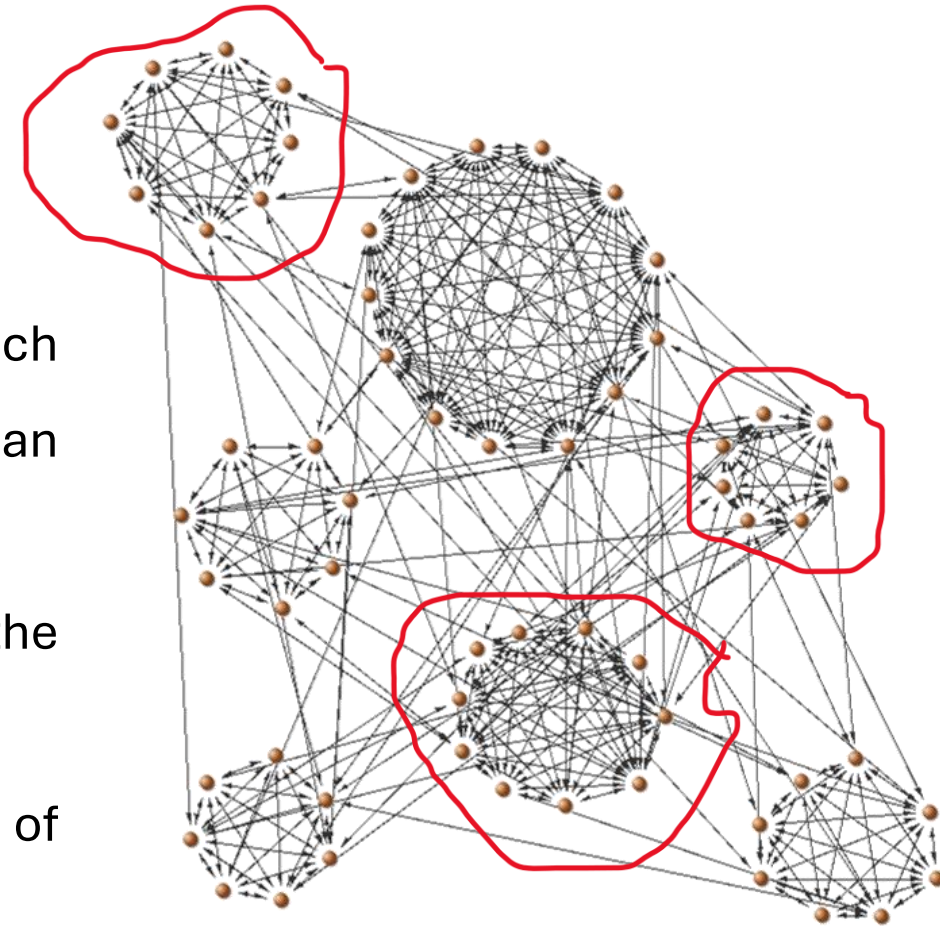
Problem: the impact of knowledge and network dynamics are difficult to disentangle.

Solution: network communities

Network communities: cohesive segments of networks that are loosely connected to other communities.

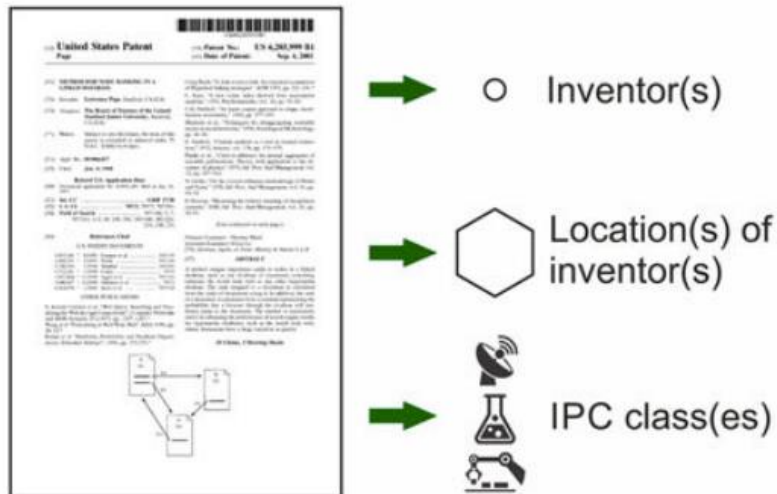
Advantages:

- Small-world networks can be decomposed to communities such that bridges are inter-community ties (Girwan and Newman 2002)
- Community detection only considers the structure of the network (Fortunato 2010) – Triadic Closure effect
- One can measure the technological specialization of communities – Technological Proximity effect



Data from European Patents

- The European Patent Office (EPO) PATSTAT database
- Creating co-inventor network in seven non-overlapping 5-year time-windows for each NUTS2 region



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Atypical patents as radical innovation

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REPORT



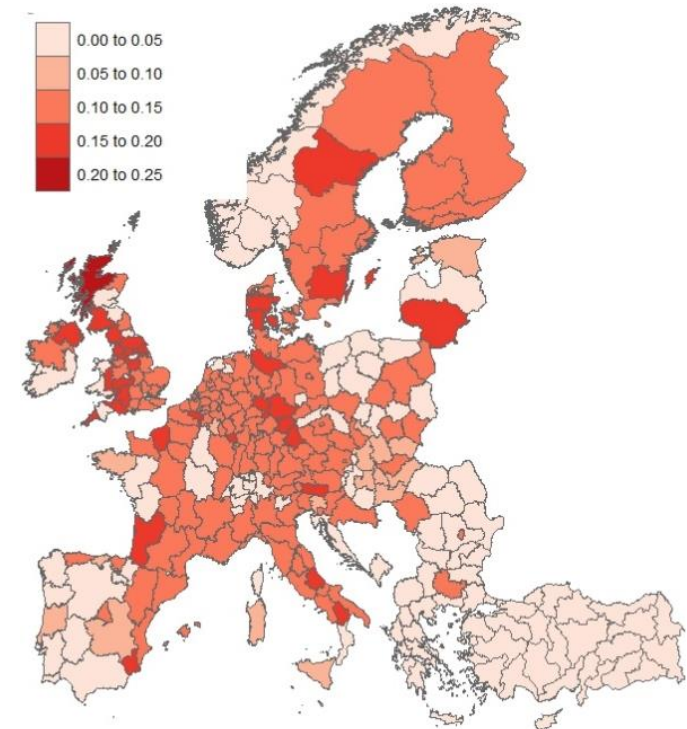
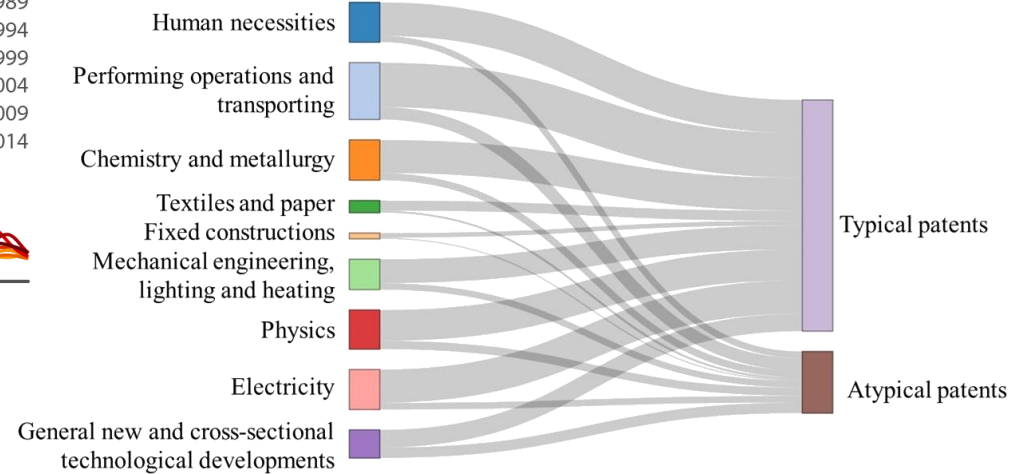
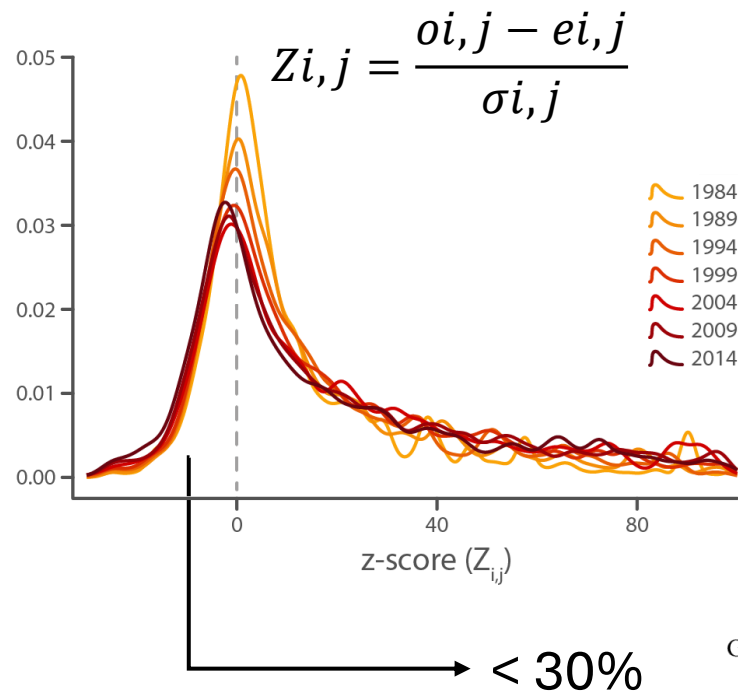
Atypical Combinations and Scientific Impact

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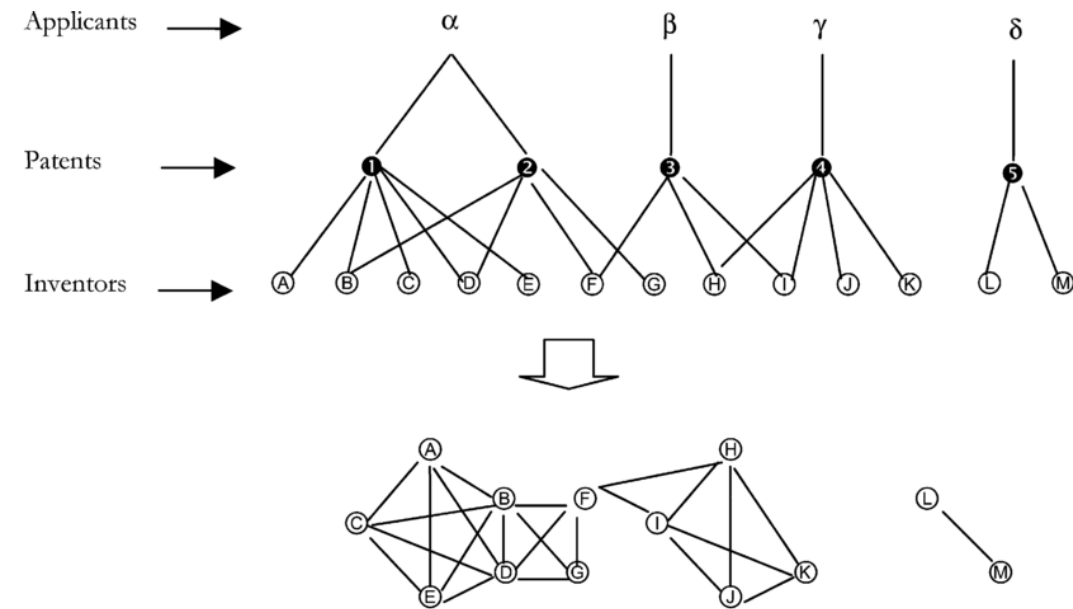
Atypical patents concentrate in cities
(Mewes 2019).



Negative values represent atypical combination of classes

Dependent variable: share of atypical patents in EU regions, to avoid the effect of scale.

Network variables 1



Network	Lattice, Ordered	Small World	Random, Disordered
Clustering Coefficient	High	High	Low
Mean Path Length	Long	Short	Short

Small-worldness (Neal, 2018)

1. Average path length / compared to a random network
2. Clustering / compared to lattice

$$\omega = \frac{L_r}{L} - \frac{C}{C_l}$$

$$SMALLWORLDNESS = 1 - |\omega|$$

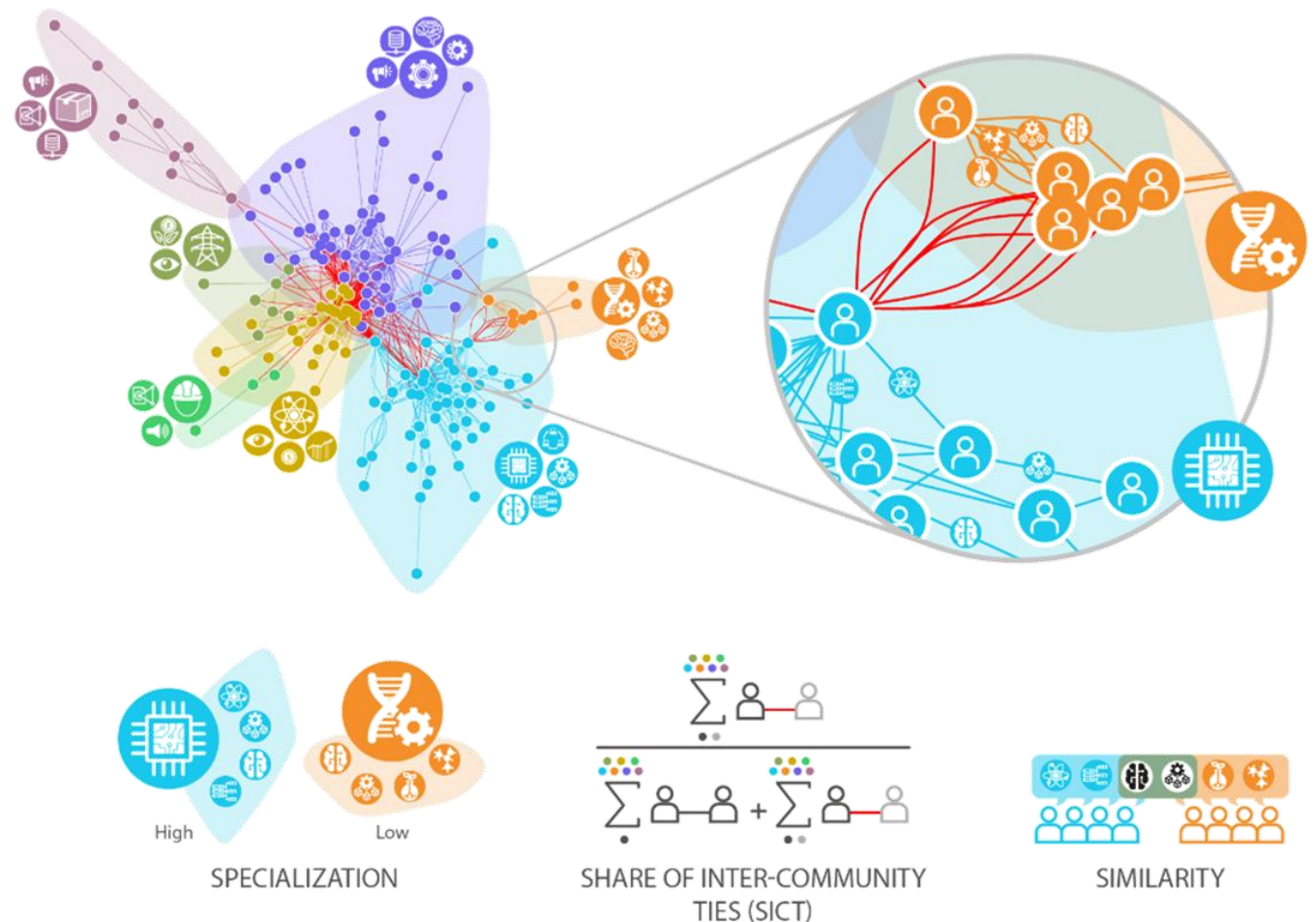
Small worldness ranges from 0 to 1.

Network variables 2

Network communities are detected by the Louvain method

Regional-level indicators

1. Specialization: median level of technology HHI in network communities (not correlated with community and inventor numbers)
2. SICT: share of inter-community ties compared to randomly rewired networks
3. Similarity: median level of technology correlations (Spearman) across connected communities



Results

$$Y_{r,t} = \alpha + \beta_1 \text{SMALLWORLDNESS}_{r,t-1} + \beta_2 \text{SPECIALIZATION}_{r,t-1} + \beta_3 \text{SICT}_{r,t-1} + \beta_4 \text{SIMILARITY}_{r,t-1} + \beta_5 N_{r,t-1} + \beta_6 Z_{r,t-1} + \varphi_t + \mu_r + \varepsilon_{r,t}$$

Controls:

1. Average Related Density (van de Wouden and Rigby 2019)
2. Complexity (Hidalgo and Hausmann 2009)
3. Density (Bergé et al., 2018; Breschi and Lenzi, 2016)
4. Isolate (Lobo and Strumsky 2008)
5. Community
6. Population
7. Interregional ties

Dependent variable: share of atypical patents

SMALLWORLDNESS	0.0115*** (0.0039)
SPECIALIZATION	0.0405** (0.0188)
SICT	0.1041** (0.0523)
SIMILARITY	-0.0342*** (0.0105)
Region FE	YES
Time FE	YES
N	1,526
R ²	0.4118